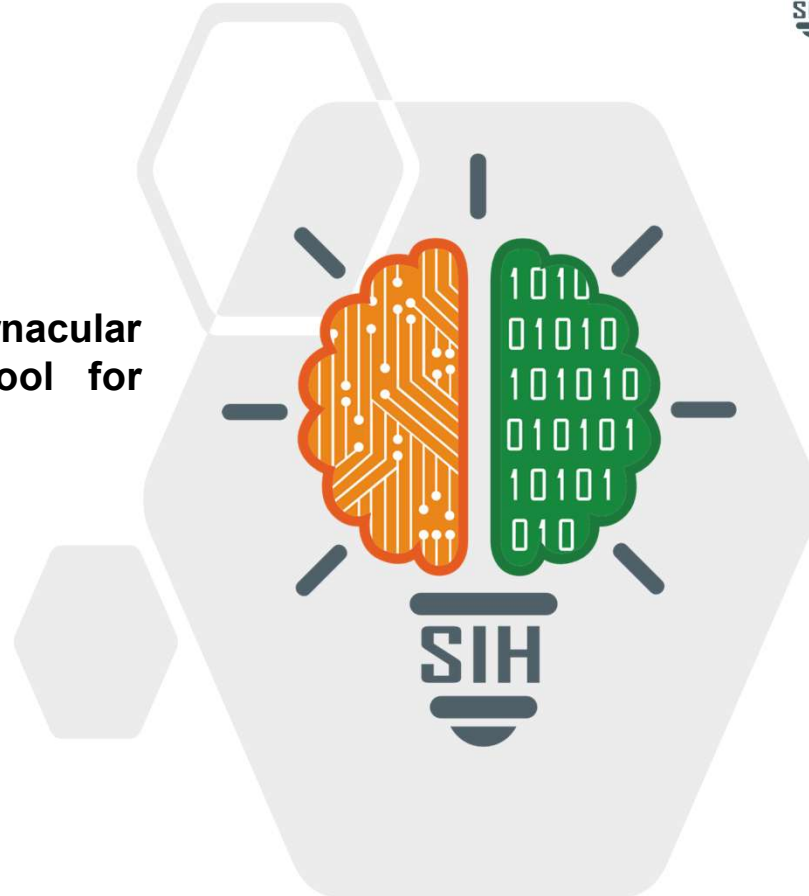


SMART INDIA HACKATHON 2026

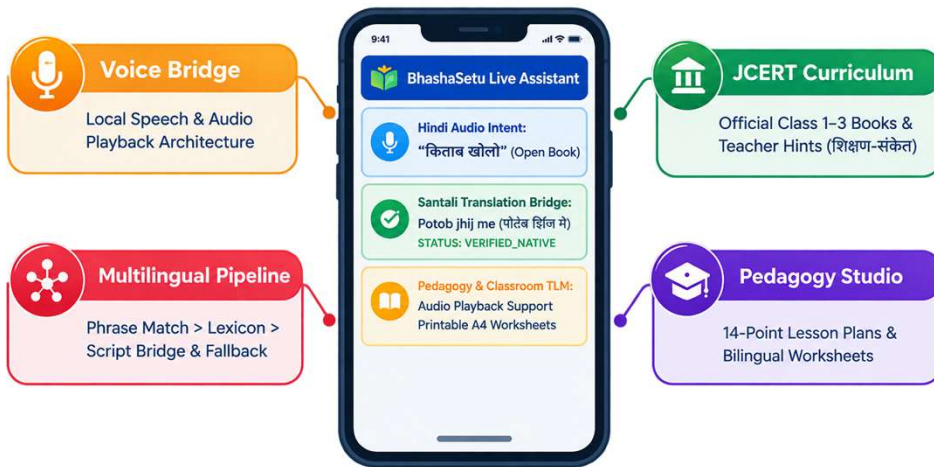


- Problem Statement ID – SIH26042
- Problem Statement Title- AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education
- Theme- Smart Education
- PS Category- Software
- Team ID- 146546
- Team Name- PROBLEM SOLVER123



❖ PROPOSED SOLUTION

- **Offline-First Multilingual Pipeline:** Hindi → Mother-Tongue support via local lexicons, phrase matching, script handling & fallbacks. Modular language packs designed for tribal languages, with coverage depending on available validated resources.
- **Teacher-First Pedagogical Studio:** Generates 14-point bilingual lesson plans grounded in tribal realia (seeds, clay, trees) and formative classroom practice.
- **Curriculum-Grounded Content Layer:** Ingests official NCERT/JCER Class 1–3 textbooks, extracting pedagogical hints (शिक्षण-संकेत) and chapter stories.
- **Voice & Multimodal Worksheets:** Local voice architecture + Web Speech API + native Ol Chiki Unicode (U+1C50–U+1C7F) & printable A4 worksheets.



➤ INNOVATION & UNIQUENESS

- **Low-Resource Quality Control:** Verified phrase → dictionary match → fallback → untranslated status tagging to prevent misleading tribal output.
- **Offline-First Architecture:** Embedded SQLite database in WAL mode stores language lexicons & textbook units locally on-device.
- **Culturally-Grounded Pedagogy:** Connects abstract FLN concepts to familiar local tribal realia (Sarjom/Sal seeds, Mahua trees, clay) for intuitive learning.
- **Community Language Validation:** Structured workflow for future native-speaker review and validation.

[01] Low-Resource NLP

Layered phrase & lexicon matching with native Ol Chiki Unicode (U+1C50–U+1C7F)

[02] Offline-First Mode

Embedded SQLite WAL mode operates reliably without continuous internet connectivity

[03] Curriculum Grounding

Indexed to official JCER & NCERT Class 1–3 textbooks & teacher hints

[04] Cultural Validation

Community review layer with strict verification status tagging

TECHNICAL APPROACH



❖ Tools & Multimodal Engine

- PyMuPDF extracts chapters, stories & teacher hints (शिक्षण-संकेत) from textbook PDFs.
- ReportLab generates bilingual A4 classroom worksheets with OI Chiki Unicode.
- Modular voice architecture extensible with local on-device speech models.

❖ Frameworks & Local Storage

- FastAPI asynchronous Python 3.11 backend with REST routing for lesson planning.
- React 19 + TypeScript + Tailwind CSS responsive PWA for school tablets & laptops.
- SQLite3 in WAL mode (Write-Ahead Logging) for ultra-fast local database operations.

CORE TECHNOLOGY STACK COMPONENTS



1. User Query / Lesson Input

Teacher speaks/types Hindi lesson intent or selects an official JCERT/NCERT chapter (Sarangi, Math, Mridang, Veena).

2. Layered Language Processing Pipeline

Hierarchical matching: Exact verified phrase lookup → Curated dictionary lexicon → Contextual rules & script translation.

3. Translation Quality & Fallback Layer

Enforces 4-tier status gating: VERIFIED_NATIVE, DICTIONARY_MATCH, FALLBACK, or UNTRANSLATED (prevents unverified output from being presented as authoritative).

4. Local DB & Curriculum Knowledge Fetch

Lightweight SQLite local retrieval fetches official teacher hints (शिक्षण-संकेत), chapter story text & cultural realia.

5. Multimodal Output Generation

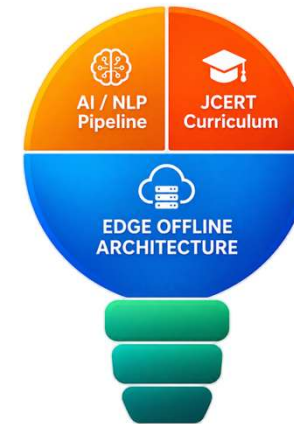
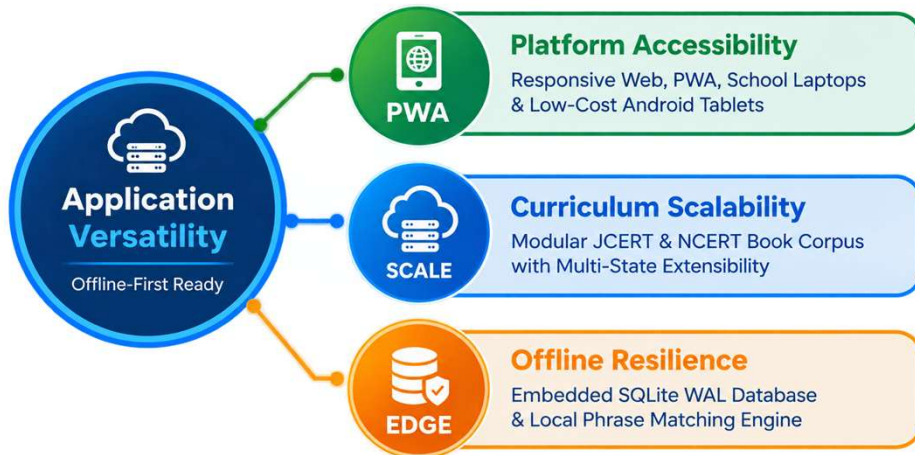
Delivers target-language text, script/pronunciation support, audio playback capability and printable worksheets.

FEASIBILITY AND VIABILITY



❖ Feasibility & Deployment:

- **Hardware Compatibility:** Runs smoothly on standard school laptops, desktop PCs, Android tablets & Raspberry Pi.
- **Offline-First Operation:** Language resources and curriculum synchronize once; operates offline during classroom teaching.
- **Extensible Language Packs:** Modular JSON/SQLite structure allows easy expansion to additional tribal dialects.
- **Low Operational Overhead:** Packages models, textbook metadata & UI locally on-device, avoiding recurring cloud bills.



Key Challenges Identified :-

Scarcity of Tribal Data

Curated mother-tongue glossaries, verified phrase dictionaries & untranslated status flags

Limited Internet

Offline-first SQLite WAL architecture + on-device asset caching (no continuous internet dependency for supported offline workflows)

Regional Dialect Variations

Community validation layer allowing local teachers & elders to verify regional terms

Phase 1: Working Prototype

React + FastAPI app, local SQLite WAL, 5 tribal packs, NCERT/JCERT ingested.

Phase 2: Language Validation & Expansion

Community validation workflows, expanded phrase dictionaries, speech playback extension.

Phase 3: Classroom Field Pilot

Teacher usability trials across rural tribal schools; measuring latency & adoption.

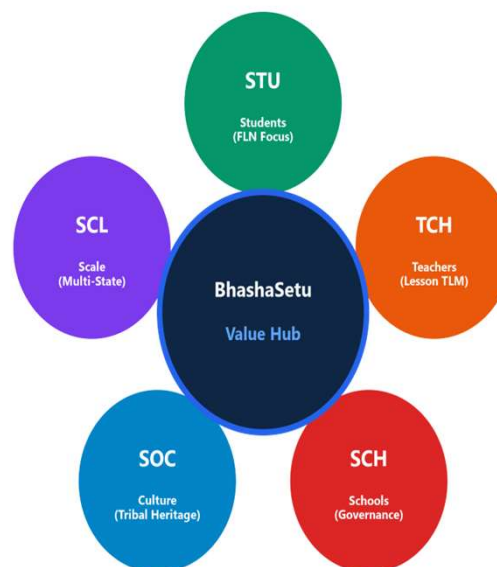
Phase 4: State & National Scale

Multi-state curriculum ingestion; integration with DIKSHA & NIPUN Bharat.

IMPACT AND BENEFITS

Impact & Benefits Breakdown

- ❖ **Students — Mother-Tongue Comprehension & FLN**
 - Mother-tongue explanations and contextual FLN practice.
 - Connects foundational concepts to familiar village realia (seeds, clay).
 - Designed to encourage classroom participation and oral-language practice.
- ❖ **Teachers — Reduced Preparation Friction & TLM**
 - Reduces repetitive manual lesson adaptation and prep time.
 - Direct access to official textbook hints (शिक्षण-संकेत).
 - Ready-to-print bilingual A4 activity worksheets for classroom use.
- ❖ **Community — Preservation & Resource Validation**
 - Structured pathway for local-language resource validation.
 - Connects home language with school learning for tribal families.
 - Promotes indigenous scripts like OI Chiki (ISO/IEC 10646).
- ❖ **Schools & Governance — Curriculum Alignment & Policy**
 - Curriculum-aligned digital learning with offline-first support.
 - Direct policy alignment with NEP 2020 (Clause 4.11) & NIPUN Bharat.



Students
Mother-tongue comprehension
Contextual FLN classroom practice

Teachers
14-Point Bilingual Lessons plans
Instant classroom voice bridge

Schools
Curriculum & NEP 2020
Alignment

Culture
Preservation of tribal scripts
Native OI Chiki Unicode (U+1C50)

Scalability
Offline-First Modular Packs
Extensible across state curricular

RESEARCH AND REFERENCES



- **NEP 2020 (Ministry of Education):**

Clause 4.11–4.13 mandates mother tongue / local language as medium of instruction in foundational grades wherever possible.

- **NIPUN Bharat Mission Guidelines:**

National mission for foundational literacy & numeracy (FLN) by Grade 3 with contextualized TLM and multilingual learning.

- **Jharkhand PALASH / MTB-MLE Framework:**

Government initiative across primary schools supporting 5 tribal languages (Santhali, Mundari, Ho, Kurukh, Kharia).

- **UNESCO & State of Education Report 2025:**

Highlights mother tongue-based multilingual education (MTB-MLE) research, UNESCO Bhasha Matters, and Jharkhand primary multilingual education context.

- **Open Tech Standards & Tools:**

FastAPI, SQLite3 (WAL mode), React 19, PyMuPDF, ReportLab, ISO/IEC 10646 (OI Chiki Unicode U+1C50–U+1C7F).

- **Live Project Prototype & Video Demo Links:**

🚀 **Live App (Vercel):** smart-india-hackathon-two.vercel.app

📺 **Demo Video:** [Watch 3-Min Walkthrough \(Google Drive\)](#)

📄 **Backend API Docs:** smart-india-hackathon-zkq9.onrender.com/docs

📁 **GitHub Repo:**

github.com/vard143/SMART_INDIA_HACKATHON

COMPARISON WITH EXISTING EDTECH & AI PLATFORMS

SIH26042 Requirement	BhashaSetu Implemented Architecture	Status
Mother-tongue learning	Local language-resource layer & realia grounding	✓ IMPLEMENTED
Vernacular translation	Local NLP + phrase/lexicon hierarchical matching	✓ IMPLEMENTED
Curriculum alignment	Official NCERT & JCERT Class 1–3 textbook parsing	✓ IMPLEMENTED
FLN learning support	14-point structured bilingual lesson planning	✓ IMPLEMENTED
Offline-first resilience	Embedded SQLite database in WAL mode & caching	✓ IMPLEMENTED
Voice interaction	Voice architecture + Web Speech API + audio playback	🟡 PARTIAL
Translation quality control	4-tier status gating (Native / Dict / Fallback)	✓ IMPLEMENTED
Low-resource tribal focus	Modular language packs with community validation	◆ EXTENSIBLE